

Spenser Millburn – Resume /Portfolio

Robotics Software Engineer / Technologist

Qualifications

Dedicated, enthusiastic and ambitious professional with excellent comprehension and retention ability. Breadth and depth knowledge in a broad spectrum of technical fields concentrated within Mechatronics/Robotics, EE, Computer Science and Mechanical Design.

CS: C++, Python, Linux, JavaScript, React.js, Bash, **ROS(1,2)**, **OpenCV**, **Matplotlib**, **NumPy**, **QT (GUI)**, **Jupyter**, Control Systems, **Forward/Inverse Kinematics**, **Gazebo Simulation**, TCP/UDP, Websockets

EE: Advanced SMT Soldering, **Altium Designer**, I2C, SPI, CAN, **Arduino/Raspberry Pi**, FPGA, Oscilloscope, Logic Analyzer, Bed of Nails, Sensors & Actuators

ME: OnShape, Creo, Solidworks, AutoCAD, Additive MFG, CNC/CAM Machining (Mill/Lathe), DFM, Fluidics/Microfluidics, Thermal Analysis

Education

Northeastern University: Mechatronics Engineering, B.S.

Class of 2022, GPA 4.0

Central New Mexico Community College: Engineering, Associate of Science

Class of 2020, GPA: 4.0

Leadership – NEU Robotics, Quadraped Robotics Project Manager & Software Lead

- Active leadership of multidisciplinary team of 75+ engineering students working to develop advanced quadraped (Robot Dog) research platform.
- Highly involved with the architectural design of the system, established a cross functional software team and spearheaded 10+ subteams.
- Substantial technical contribution to the development of CAPSTAN servo reduction mechanisms, ROS (Robot Operating System), Simultaneous Localization and Mapping, Computer Vision, Inverse kinematics and more.

Practical Experience

(March 2022- Current) **Robotics Engineer** – Building Machines Incorporated

- Full ownership and architecture of a high powered (480V 50Hp) robotic fluid delivery system for additive manufacturing.
- Designed implemented tested and integrated a full stack React GUI to override physical control panel.
 - WebSocket interface and Flask python backend. Systemd linux service management.
- IOT integration of multiple I2C and SPI sensors/Actuators with custom PCBs and C++ drivers.
 - Modalities: Laser Time-Of-Flight, Depth CV, Pressure Transducers, Solenoid Valves, BLDC motor, Relays
 - Data acquisition, analysis and visualization with JupyterLab and Python.
- Electromechanical design, build, test of fixtures

(March 2021- March2022) **Engineering Technician** – Owl Labs

- Rapid prototyping, prototype assembly and design. Supported 30+ FTEs.
- Professional fixture design, fabrication, testing and deployment. Contract manufacturing production line integration (10000+ units annual throughput)
- Thermal data acquisition and Test Engineering.
- Customer facing Embedded Linux and Android software development support.
- Data acquisition, analysis and visualization not limited to thermals, acoustics and airflow.

(August 2020- March 2021) **Engineering Technician** – Degree Controls Systems

- Provide SMT rework and circuit analysis support.
- Project management of Engineering Change Order and Prototype Change Order documentation.
- Design PCBAs and electro-mechanical assemblies.
- Invented customer facing pythonic software, GUI design with PyQt5.
- Provide mechanical engineering CAD support of fixtures and jigs

(August 2018 – August 2020) Operations Technician (Engineering Focused) – Sartorius NA

Note: See portfolio appended to the bottom of this document

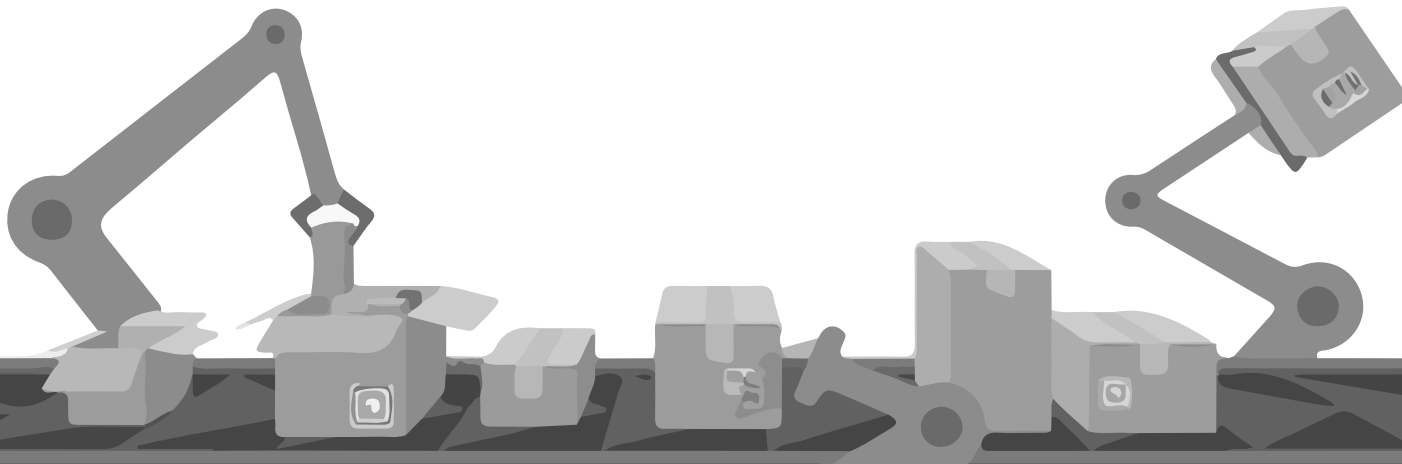
(August 2018 – August 2020) Operations Technician (Engineering Focused) – Sartorius NA
Design, fabricate and document professional production jigs, ultimately generating a return greater than USD \$50,000 in automated labor and recovered inventory.

- Designed, prototyped and established custom IOT automation server framework, developed exclusively in python3
 - Automates paperwork system, specifically targeting key KPI and Metric calculations.
 - Utilizes custom microcontroller clients to report live production data via a socket server.
 - Automates label printing and generation.
 - Provides a dynamic project management platform, in order to achieve better insight and communication to managers via the server's web based GUI
- Successfully modeled and presented a restructured production floorplan to executive level administration
- One of three technicians to Manufacture, troubleshoot and certify highly accomplished equipment (specifically flow cytometry) (Unit price: \$330k USD)

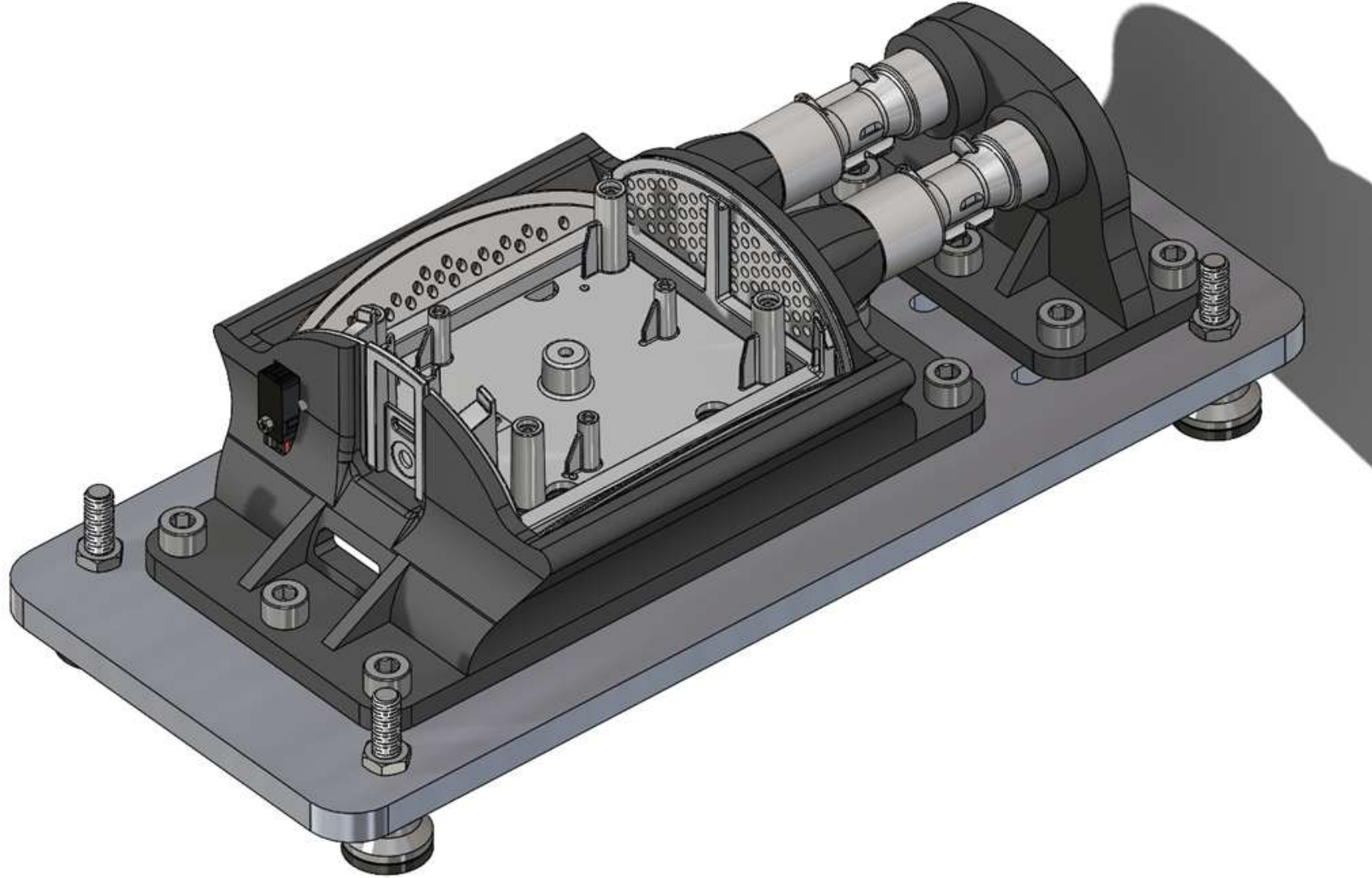
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Process/Industrial Engineering

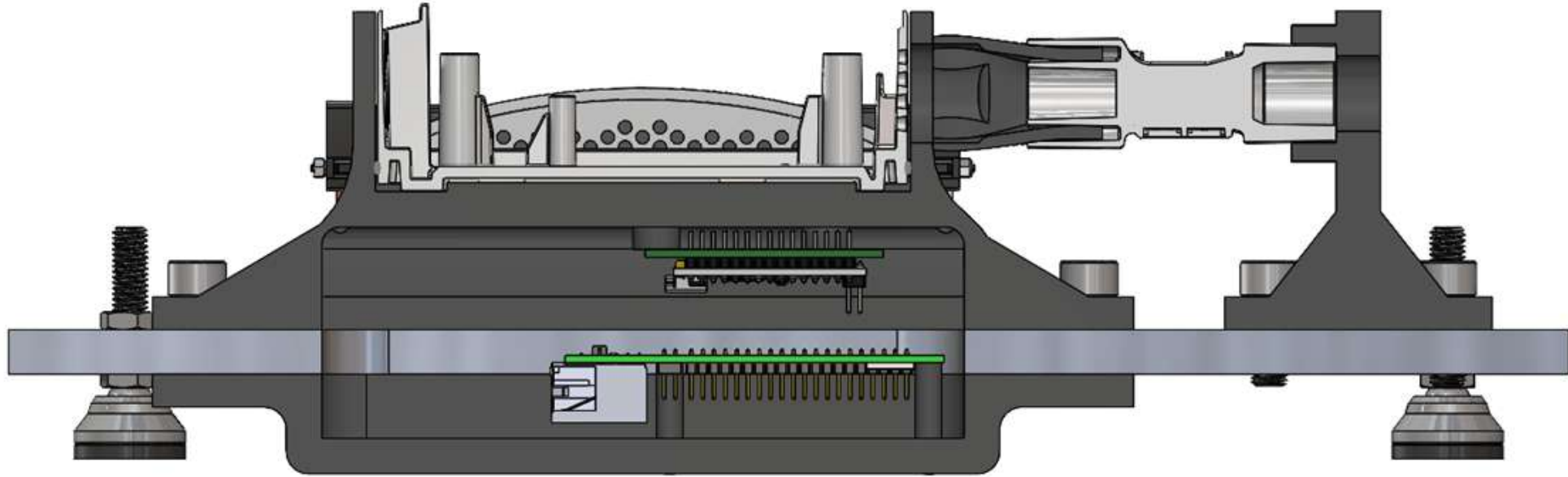
Spenser Millburn – Portfolio 2019-2021



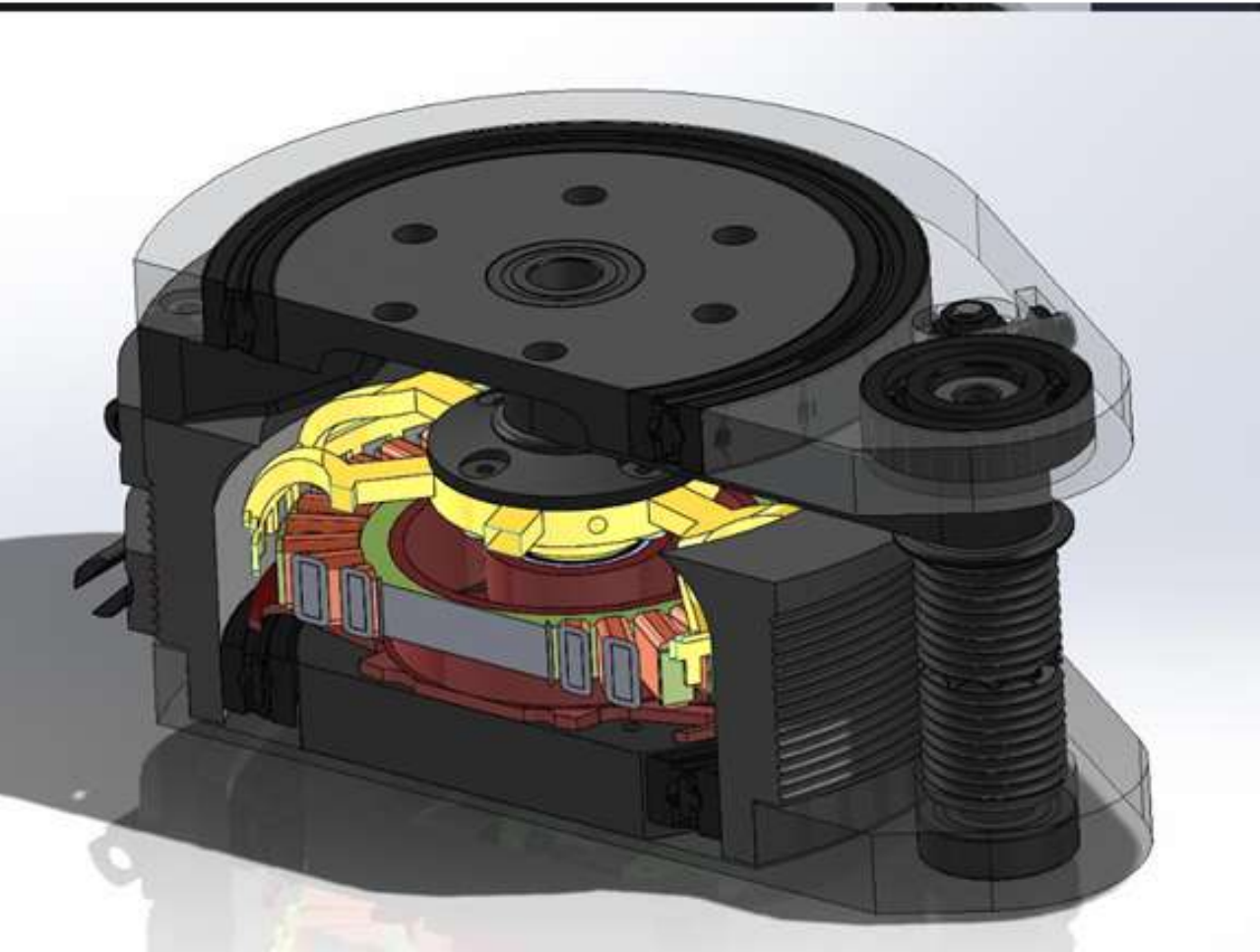
Production Fixture: High Throughput Airflow Validation



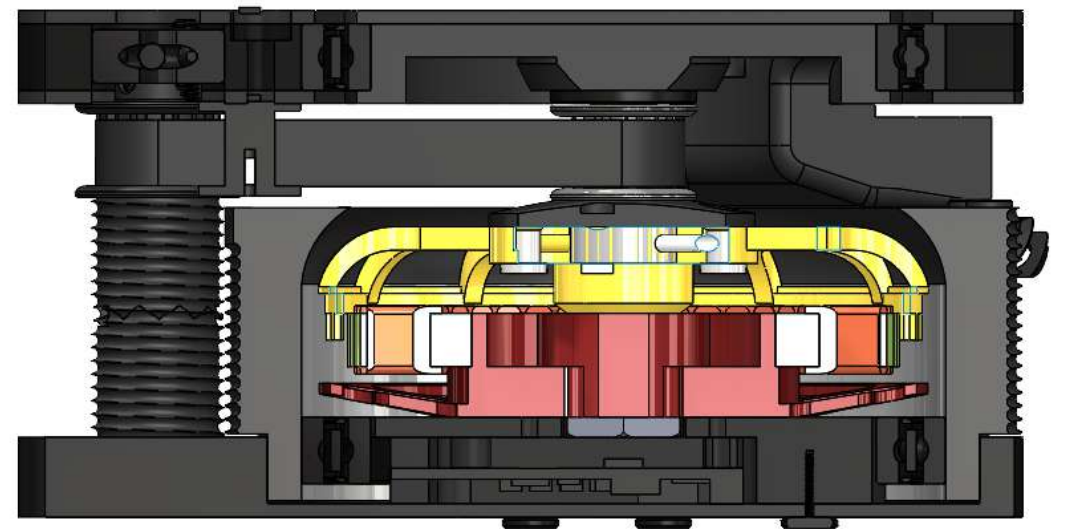
Production Fixture: High Throughput Airflow Validation



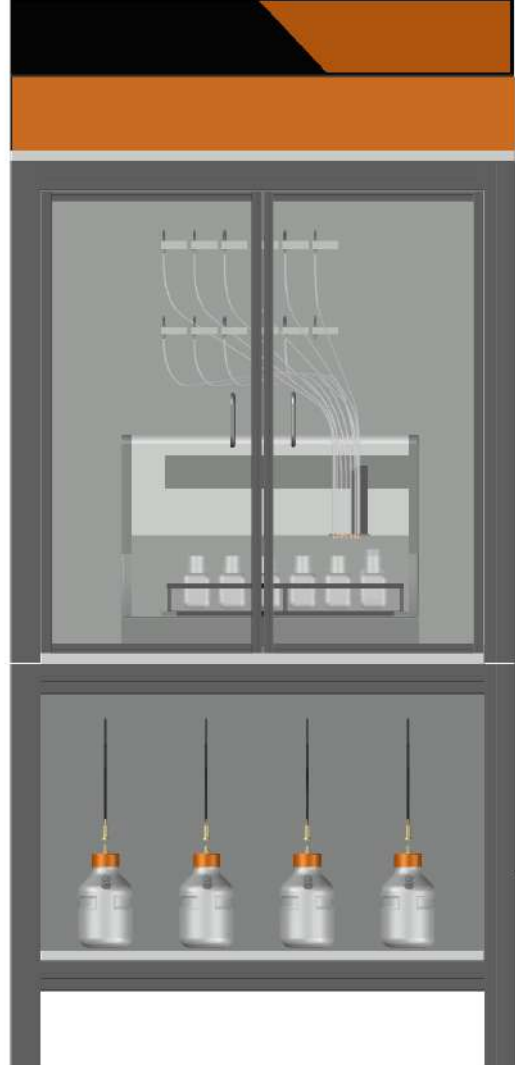
Quadruped Robot: Capstan Drive



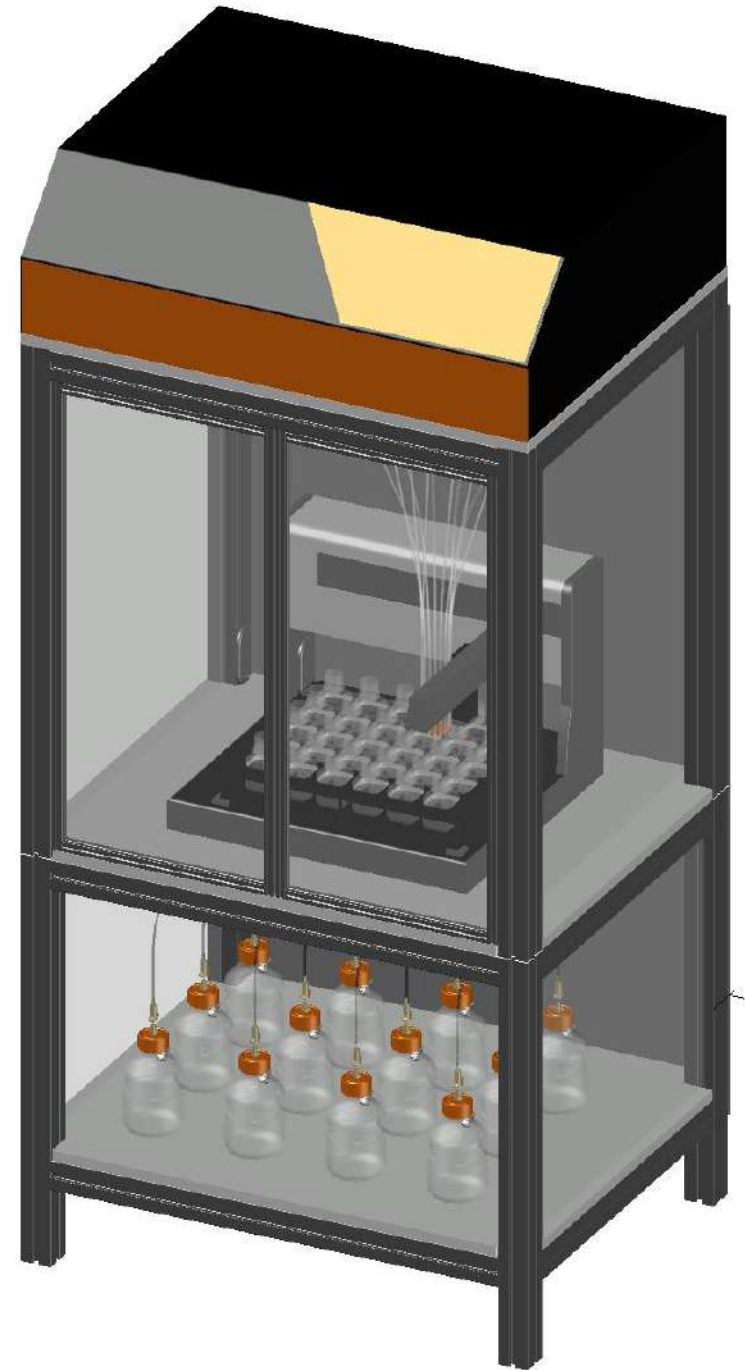
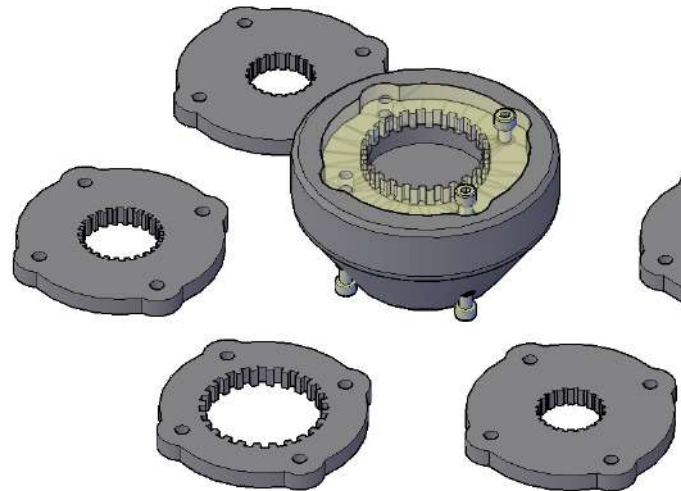
- Cable driven gear reduction
- Brushless motor successfully mounted internally in the large drum.
- Less than \$200 in parts
- Parametric model in process for easy reconfiguration of gear ratios.



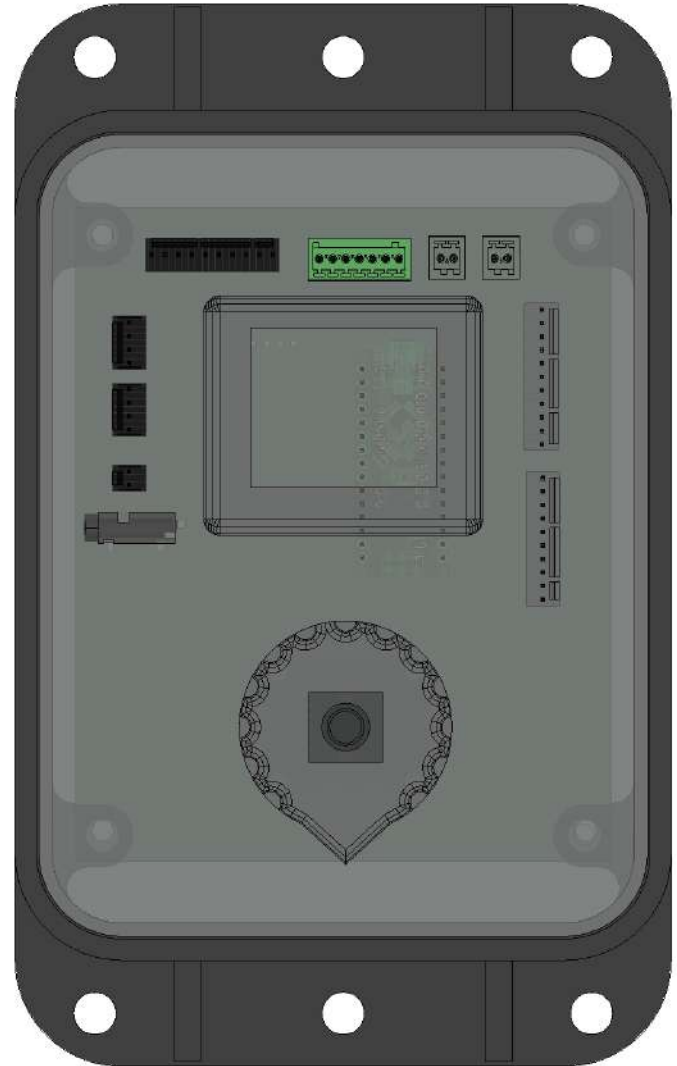
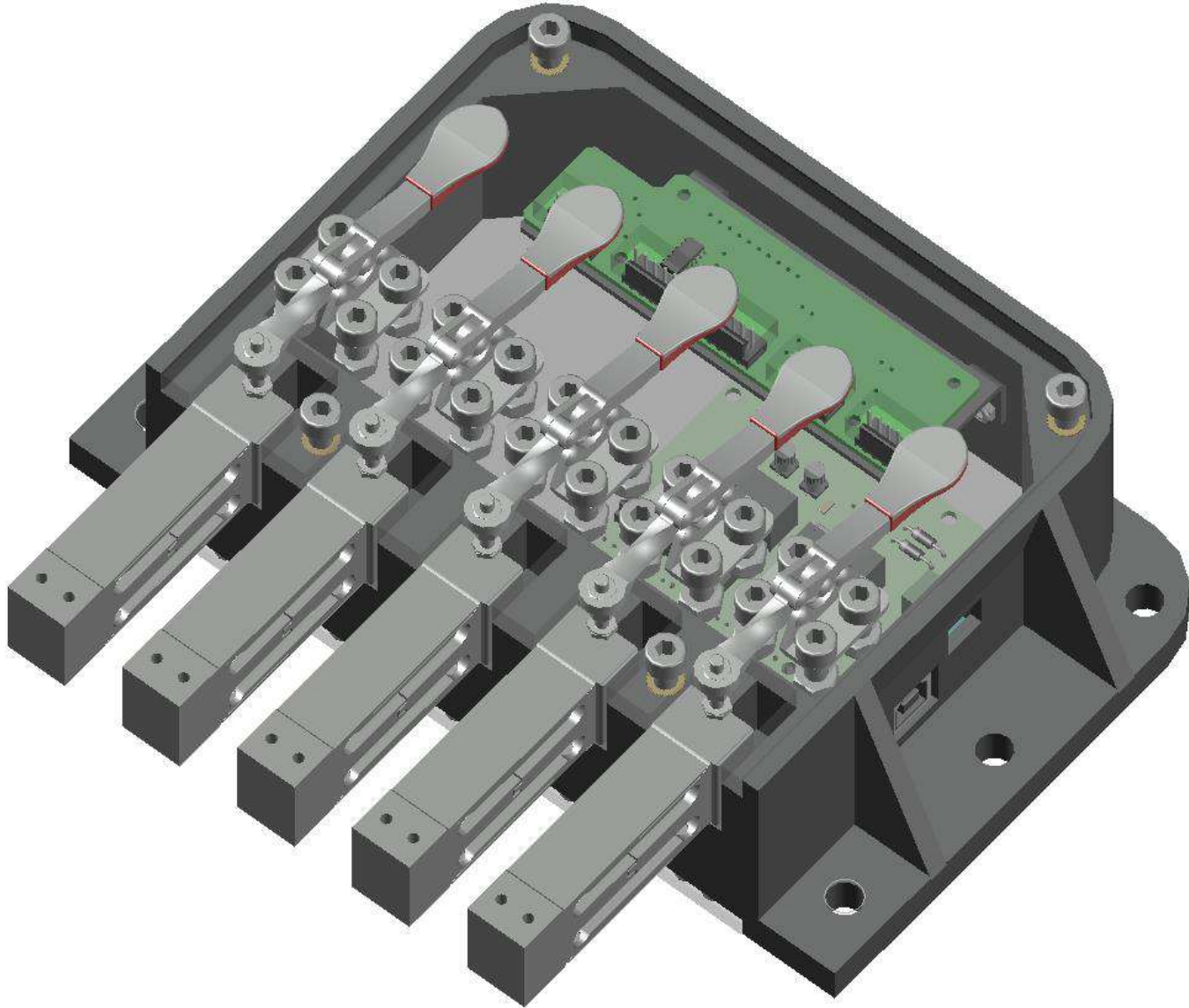
Reagent Lab Automation: Liquid Handler/Capper



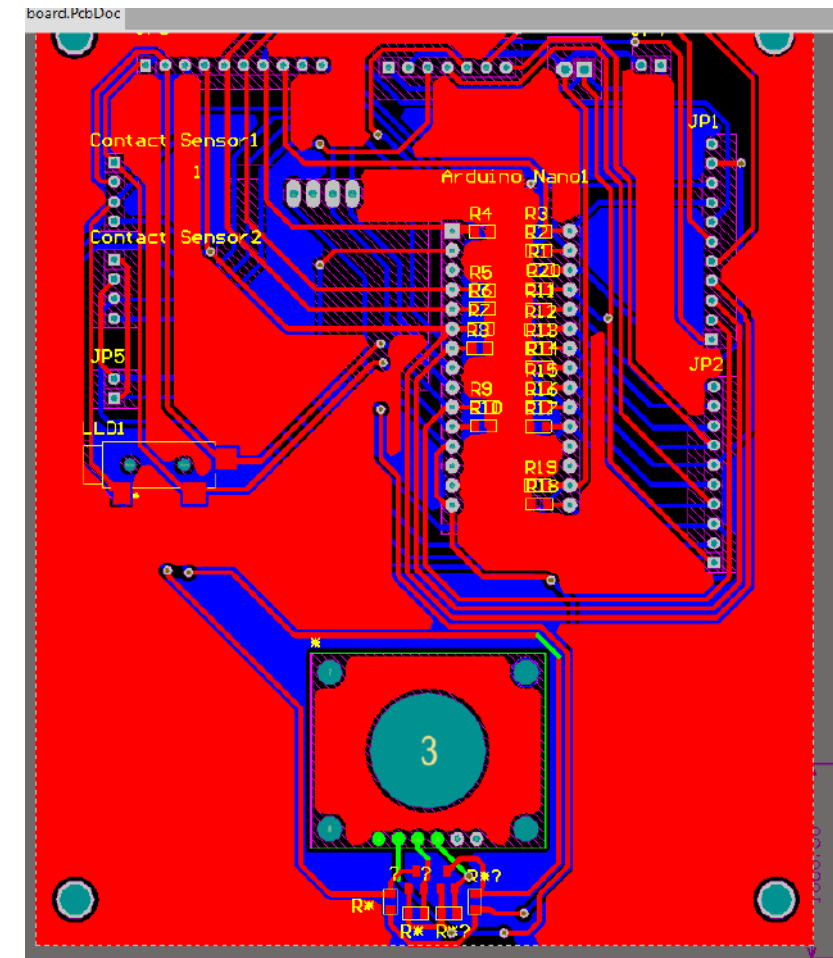
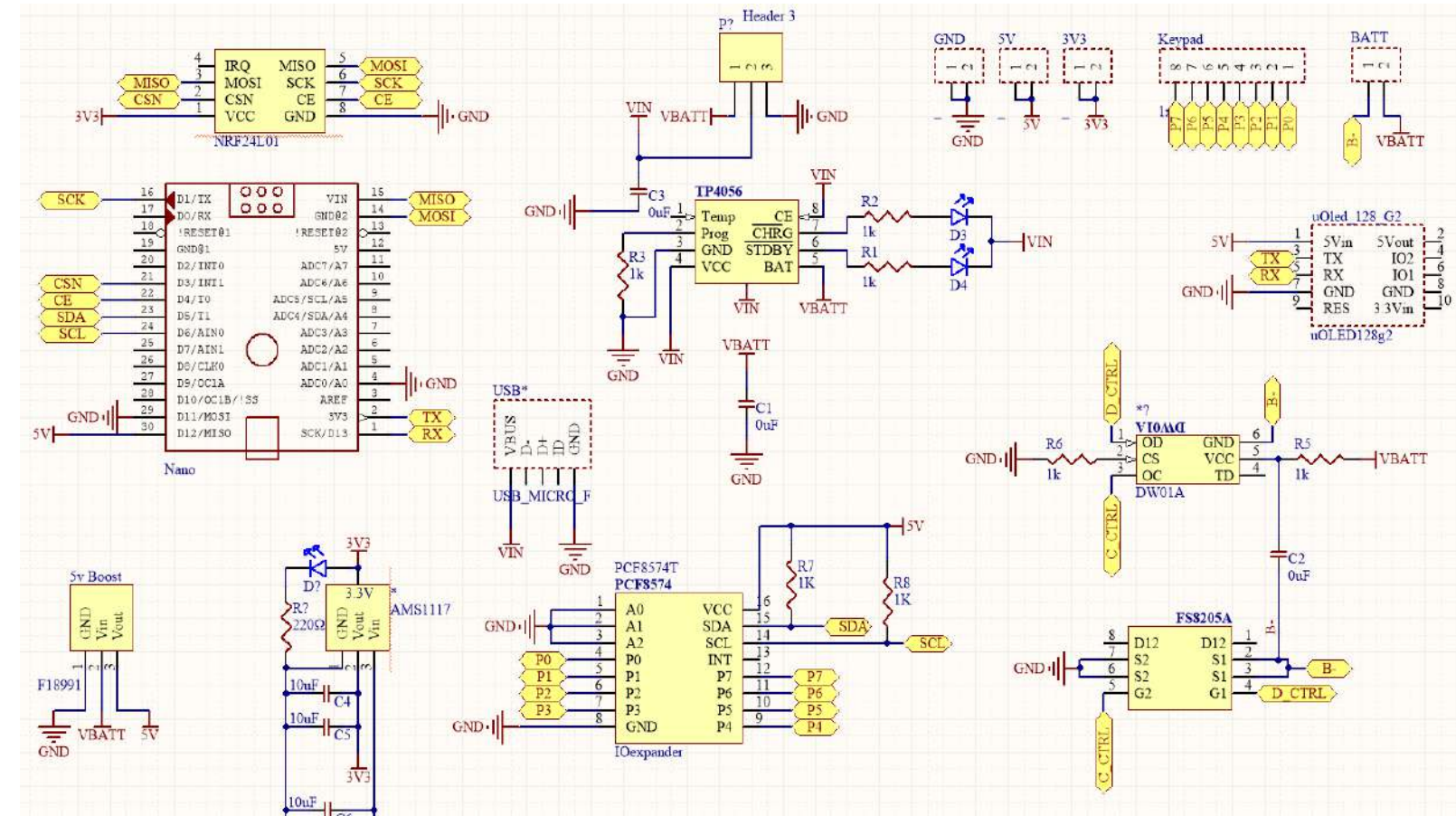
- Based on GX 271 liquid handler
- CO2 dispensing system
 - **Solenoid Valves** controlled by microcontroller/microprocessor
- Custom capping tool
- Laser cut trays to secure bottles



Hardware: Test Stations



Cable Test Station : PCB Design

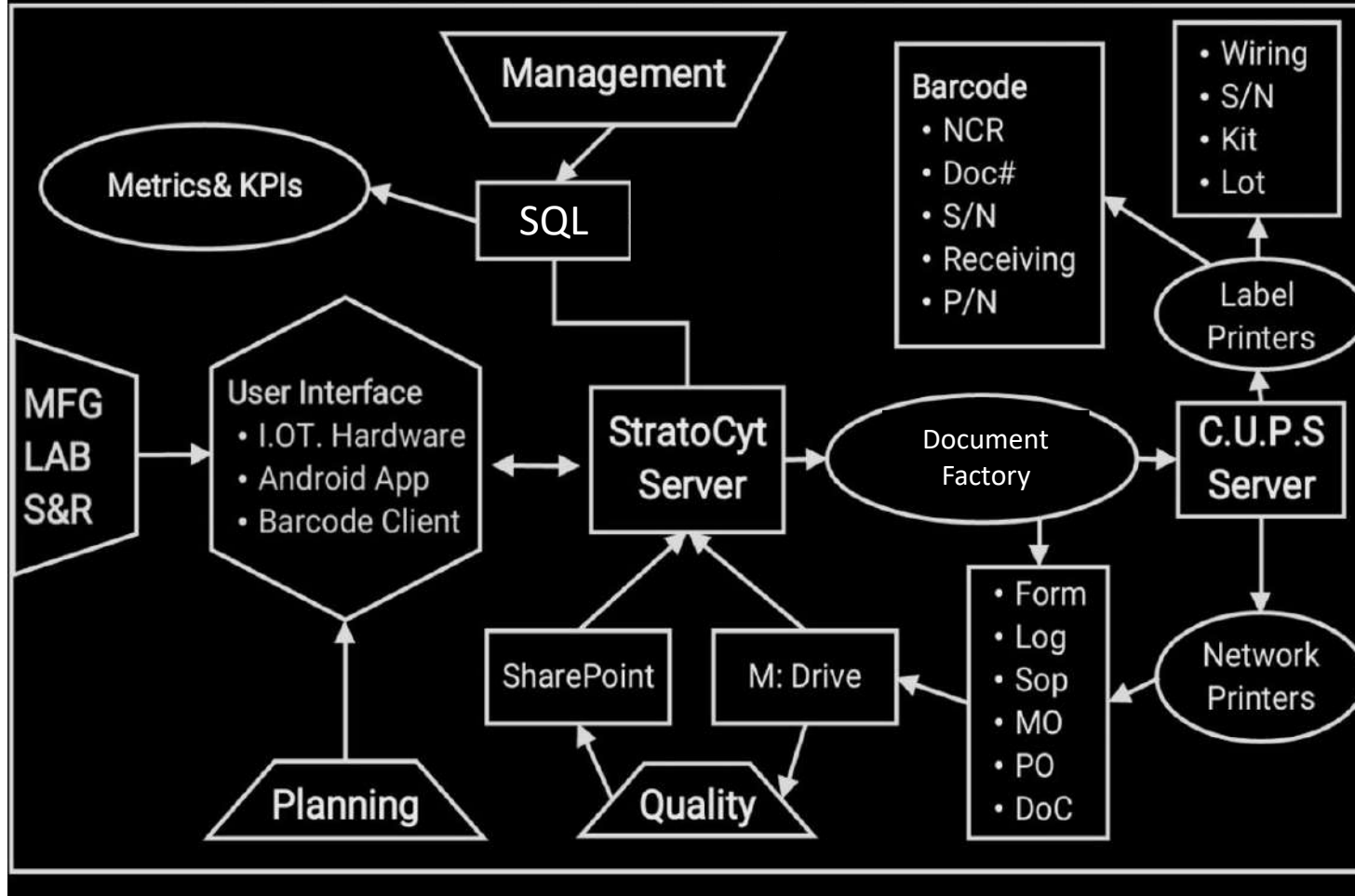


PCB : Rapid Prototyping, Hand Etching



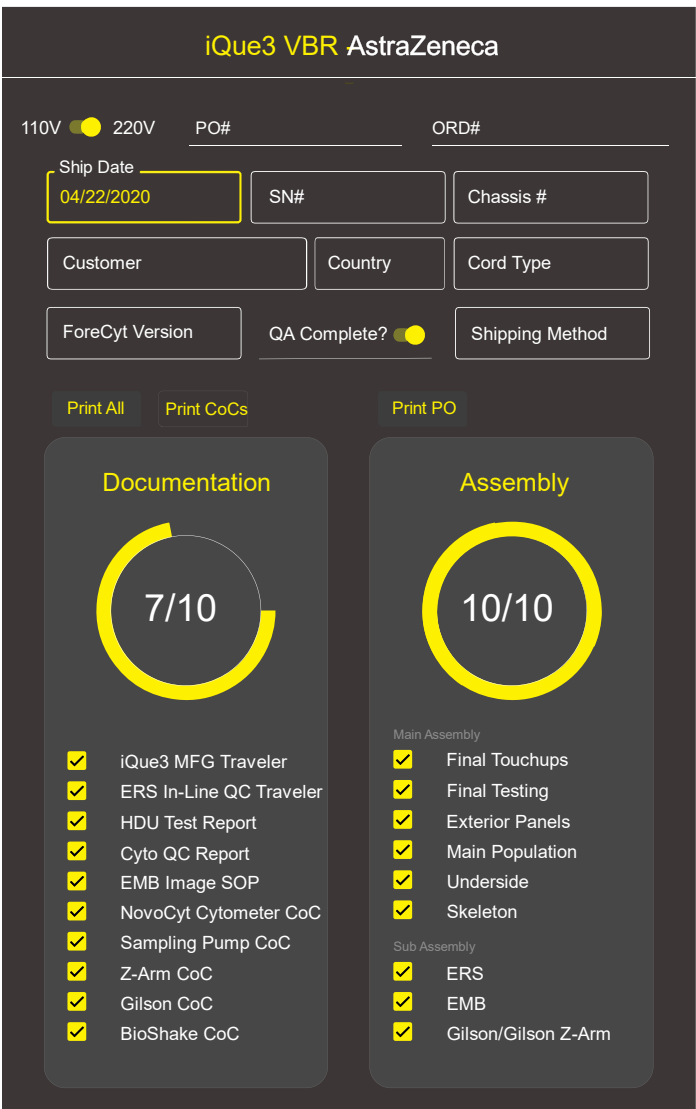
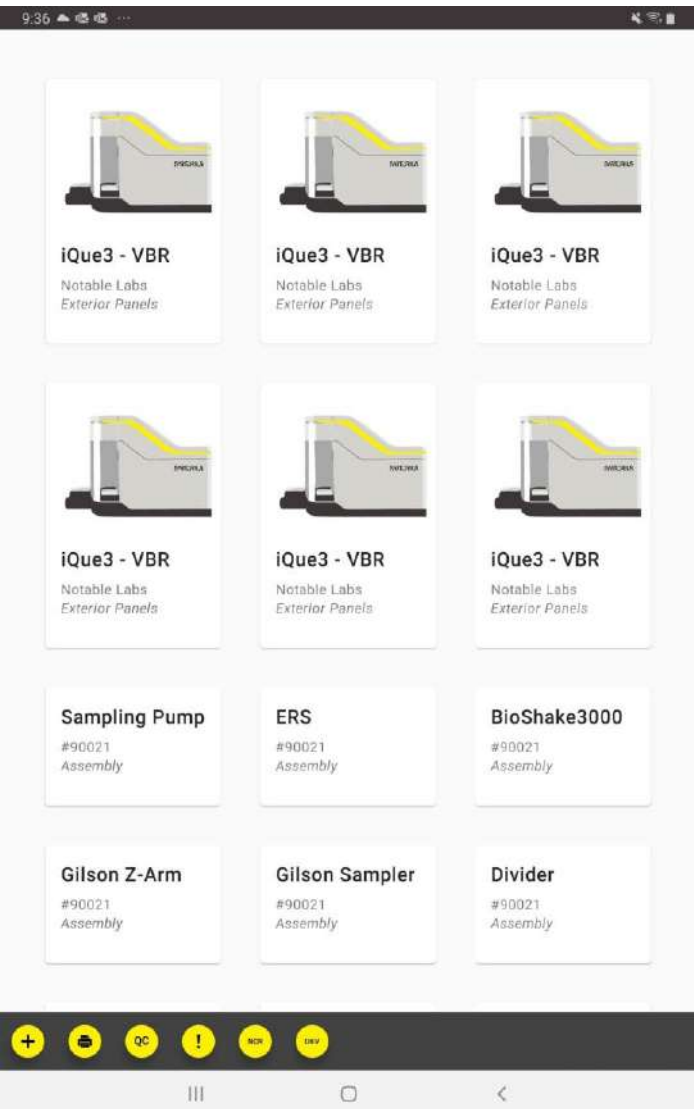
StratoCyt – Custom Automation Server

Reference Diagram



- Microsoft SQL Server Database
 - Stores essential production data for real time metrics
- Python Socket Server
- Automatic document and label generation
 - Automates paperwork system
- Mobile front-end for efficient in process user interface

StratoCyt: Front end: Android app

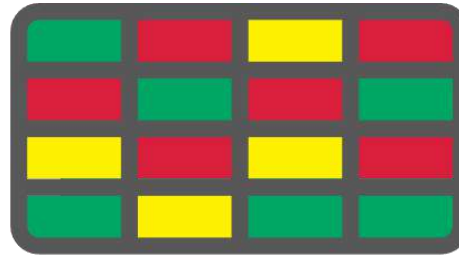


Pick by Light System

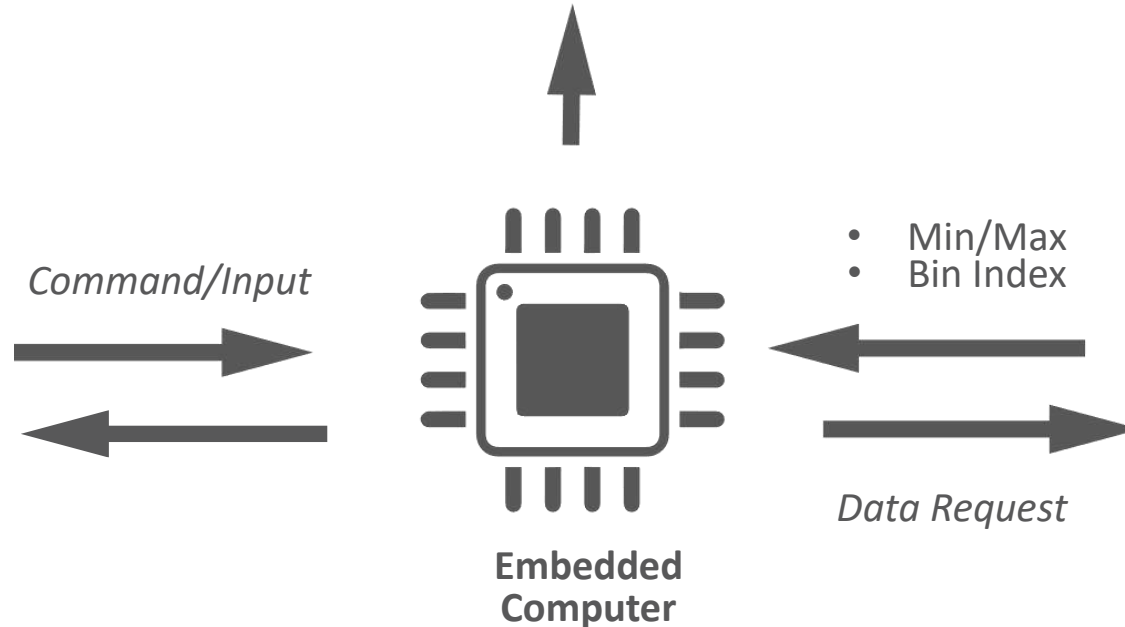
Uniquely Illuminable Component Storage

- Less **time** spent pulling inventory, more during assembly
- Easily avoid **shortages** via the Kanban system

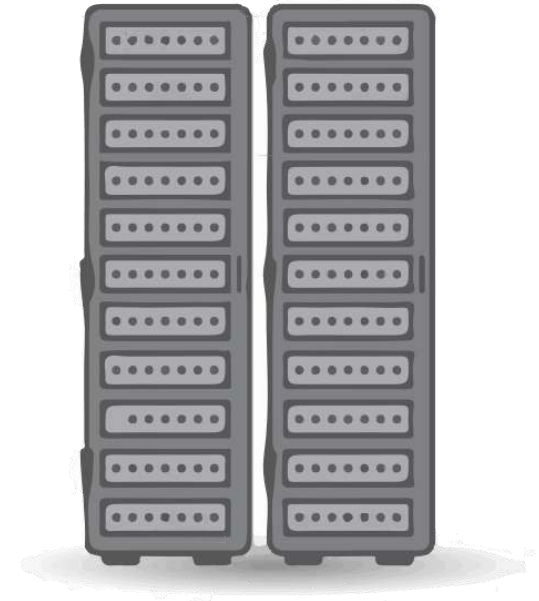
LED Manifestation of Location/Level Intuitive Pulling of Inventory



Android UI
(StratoCyt)

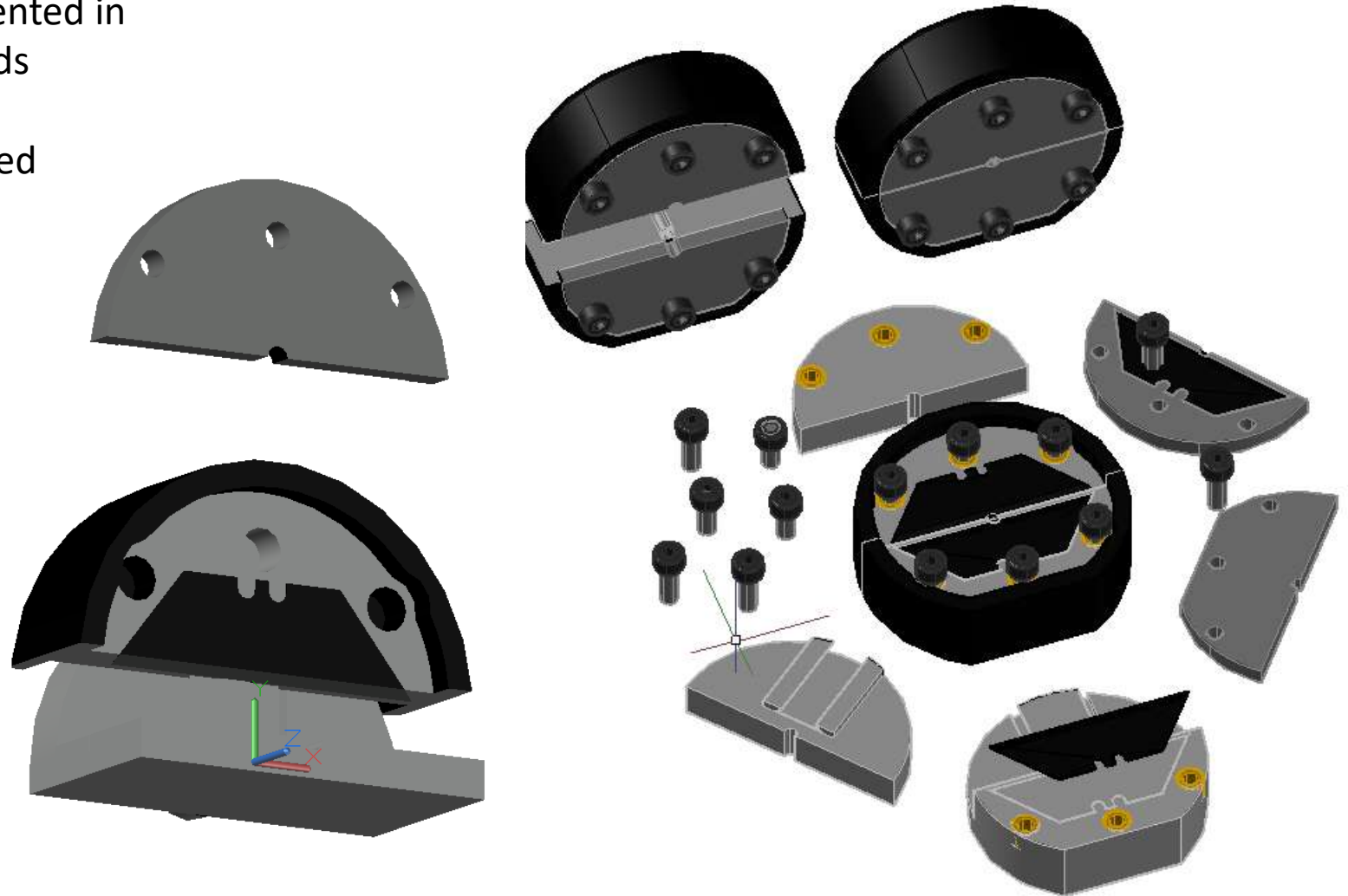


SQL Database

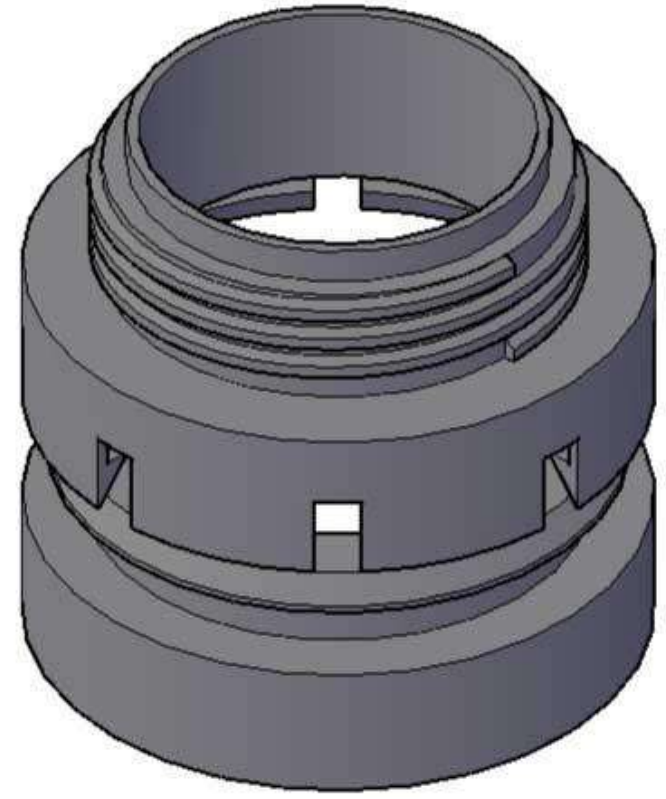
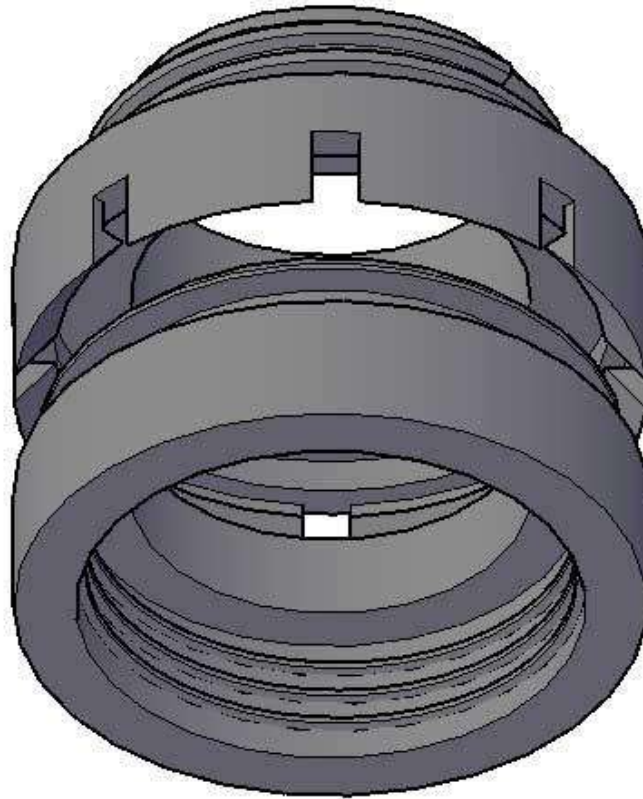
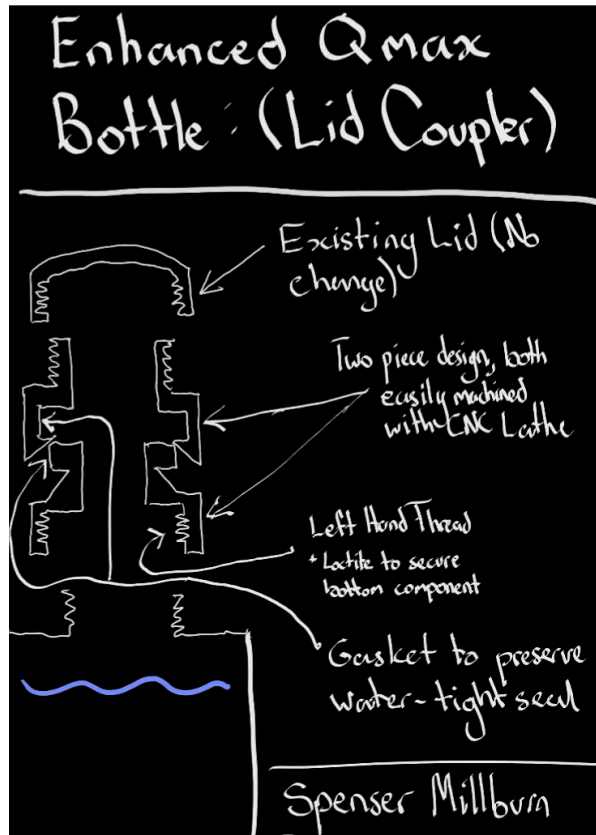


Microfluidics: Rework Jig

- Designed, fabricated, tested and implemented in 72hr window to meet production demands
- Effectively recovered USD \$20k in reworked material due to out of warranty vendor error



Qmax – Fluidics/Electrical Bearing`



Load Cell Calibration Jig

